Hanoi Urban Noise Mapping

Data Collection Report

323 smartphone measurements · 3 sites · 10 Jun to 14 Jul 2026

323	3	47-88	0.45	39%
Measurements	Sites	dB range	Model R^2	Exceed QCVN

1. Objective

Map and characterise urban noise in three contrasting districts of Hanoi, and predict noise from urban morphology. We follow the Sunbird AI methodology (Urban Noise Uganda 61K, Nsumba et al., 2026): smartphone field collection plus a morphology-to-noise model, reproduced then applied here.

2. Data collection summary

Site	n	Median dB	Min-Max	% < 60 dB	% roadside
Ocean Park	184	68	47-88	28%	73%
Hoan Kiem lake	79	71	55-80	9%	86%
Vinh Tuy area	60	64	53-76	27%	65%
ALL	**323**	**68**	**47-88**	**23%**	**75%**

Zones: Ocean Park = new high-rise development (shielding, construction); Vinh Tuy = transport infrastructure (heavy traffic); Hoan Kiem = historic old quarter (narrow streets, pedestrian zones).
QC: site labels cross-checked against GPS (6 mislabelled submissions reassigned).

Headline findings: median 68 dB, 39% of measurements exceed the QCVN limit, noise peaks at 18:00. A model trained directly on our data reaches R^2 0.45 (honest cross-validation, section 5).

3. Methodology

3.1 Field data-collection protocol

Study zones were mapped in Google Earth and chosen for contrasting typologies (section 2). At each point we recorded a 20-30 s reading (with a 10 s audio clip) in ODK Collect, submitted live to a KoboToolbox server with automatic timestamp and GPS. Phones were held at 1.2 m (SPB positioning) at varied distances from the road, by three cross-calibrated collectors, from 05:00 to 23:00 with emphasis on rush hours. Weather (Open-Meteo) and spatial context (OSM) are added.

3.2 Reproduction of the Sunbird pipeline

We reproduced the full Sunbird chain on their Uganda dataset (notebooks 01-06): cleaning, audio QC, morphology features, figures, surrogate model. Our statistics match the paper (median 49 dB vs their 45-50 dB).

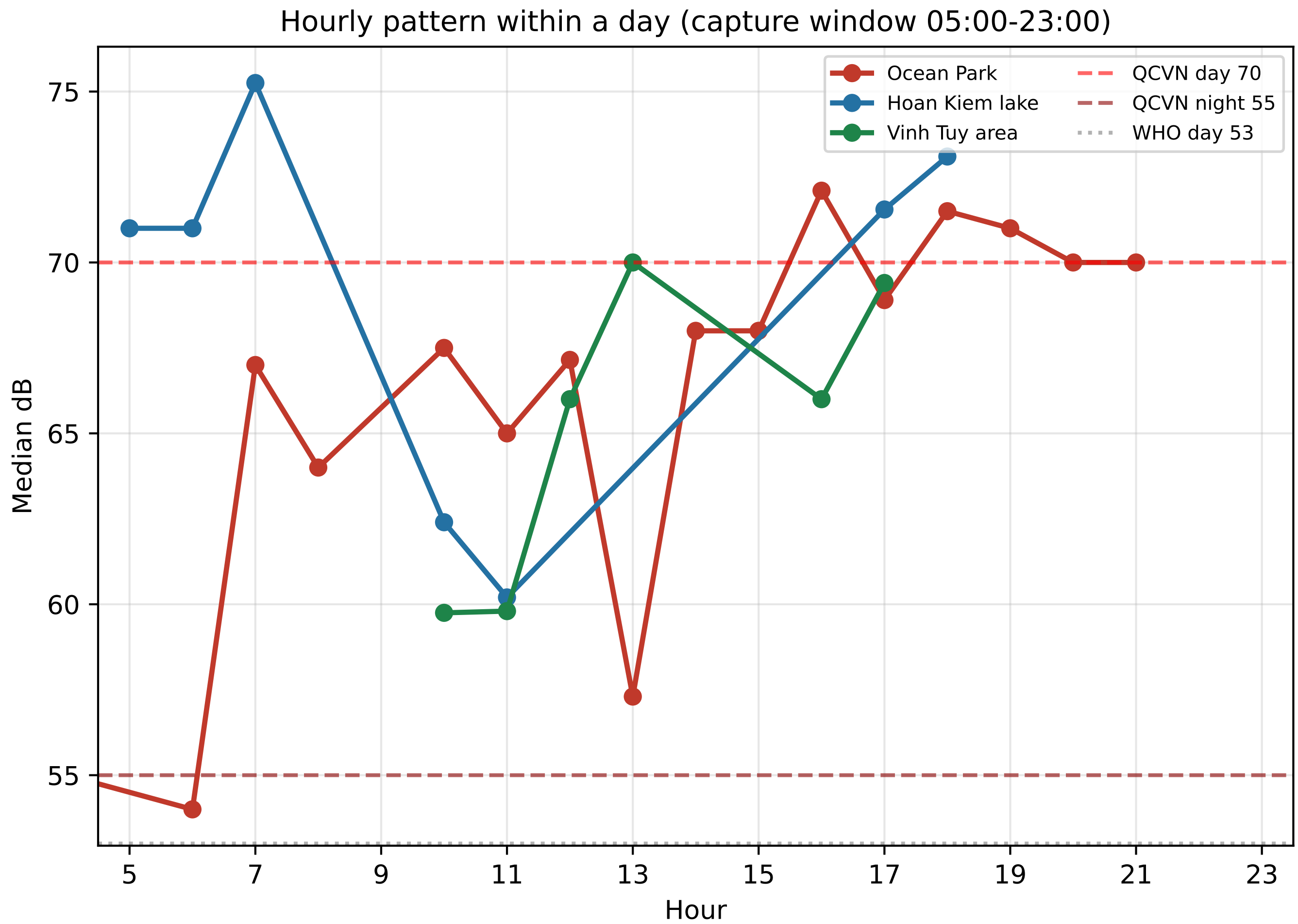
3.3 Model features

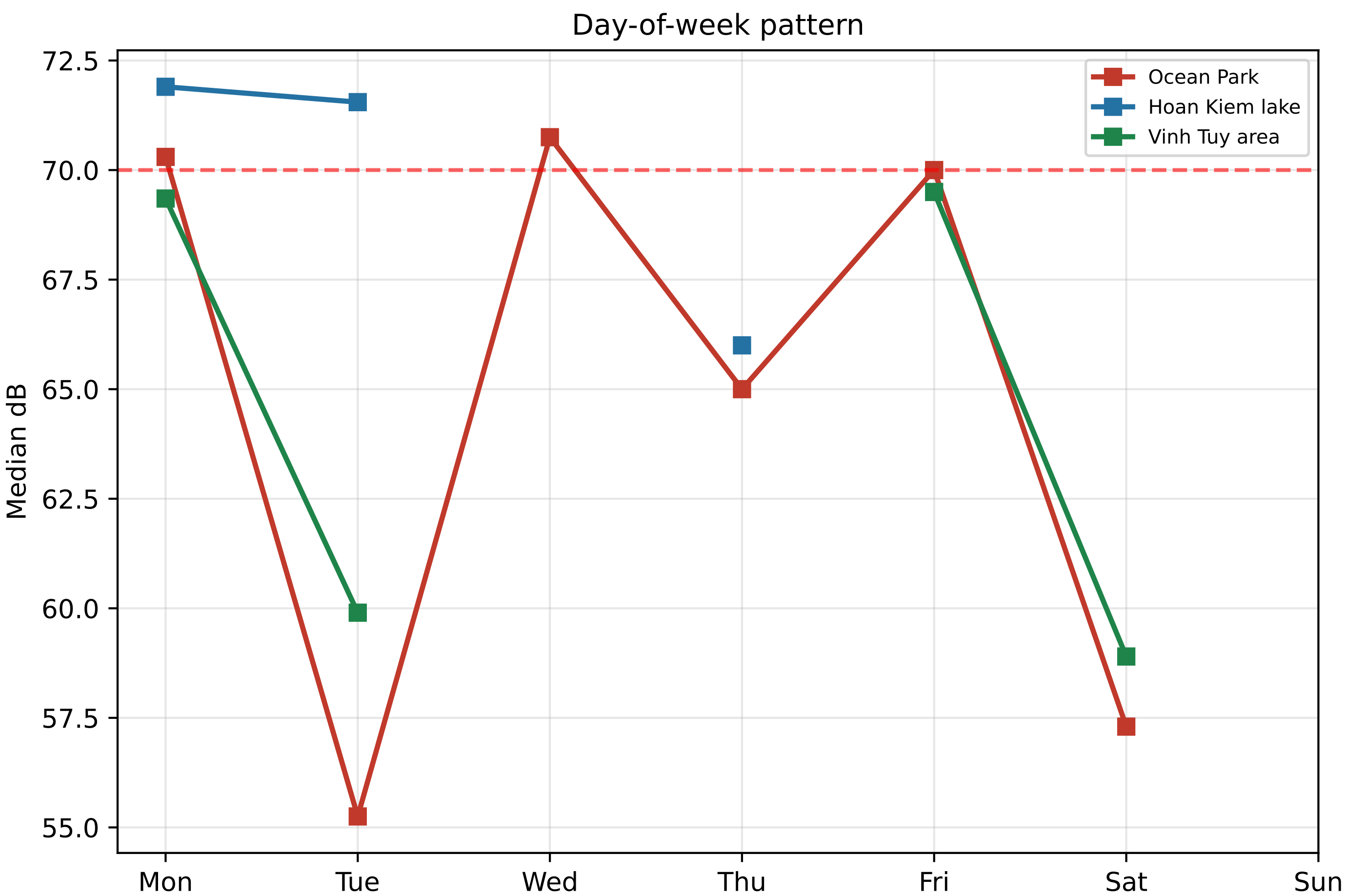
Morphology within 300 m of each point (OpenStreetMap): built-area ratio, road density, intersection count, distance to nearest road; plus hour of day and weekend flag. Model: LightGBM.

3.4 Standards used

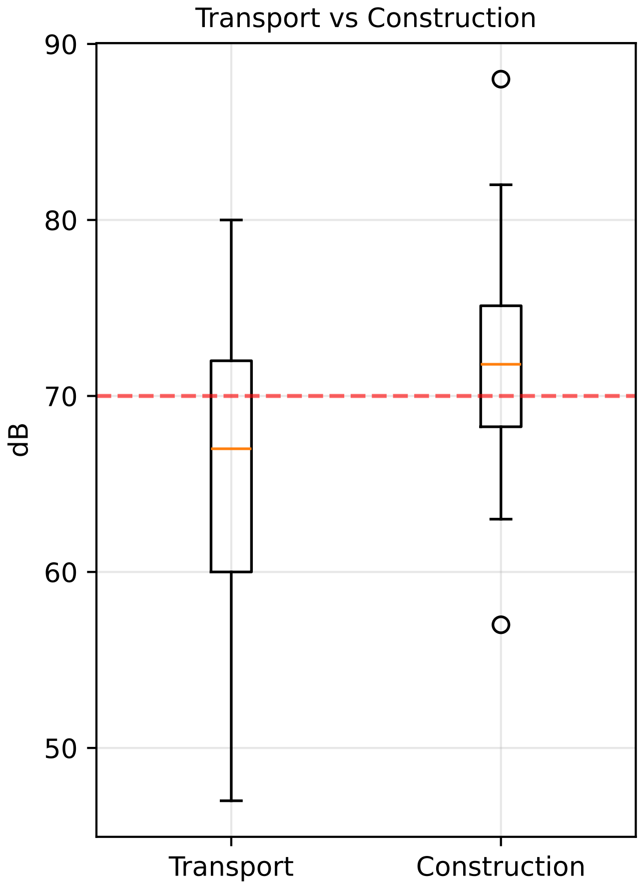
Standard	Day (6-21h)	Night (21-6h)
QCVN 26:2010/BTNMT (Vietnam, ordinary area)	70 dB	55 dB
WHO (road-traffic guideline)	53 dB	45 dB

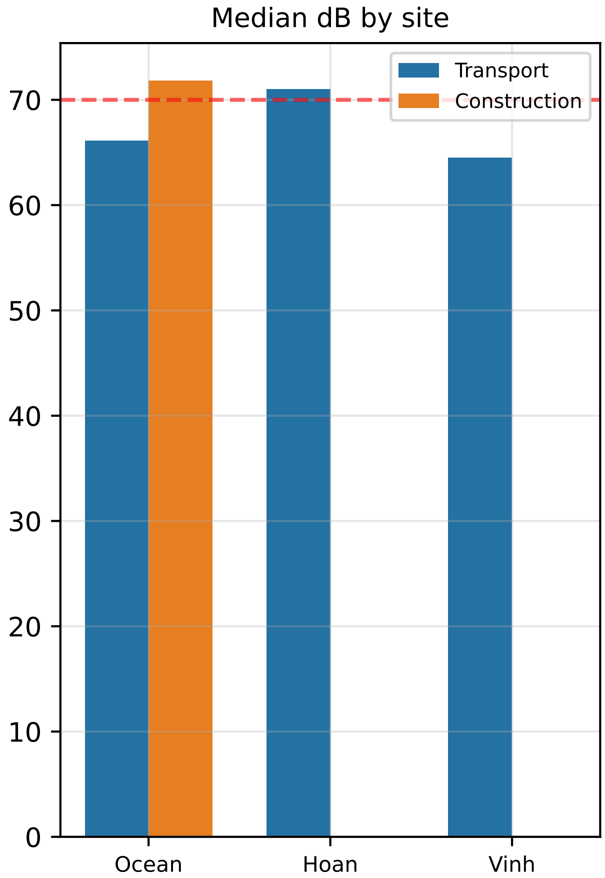
4. Data analysis: temporal patterns

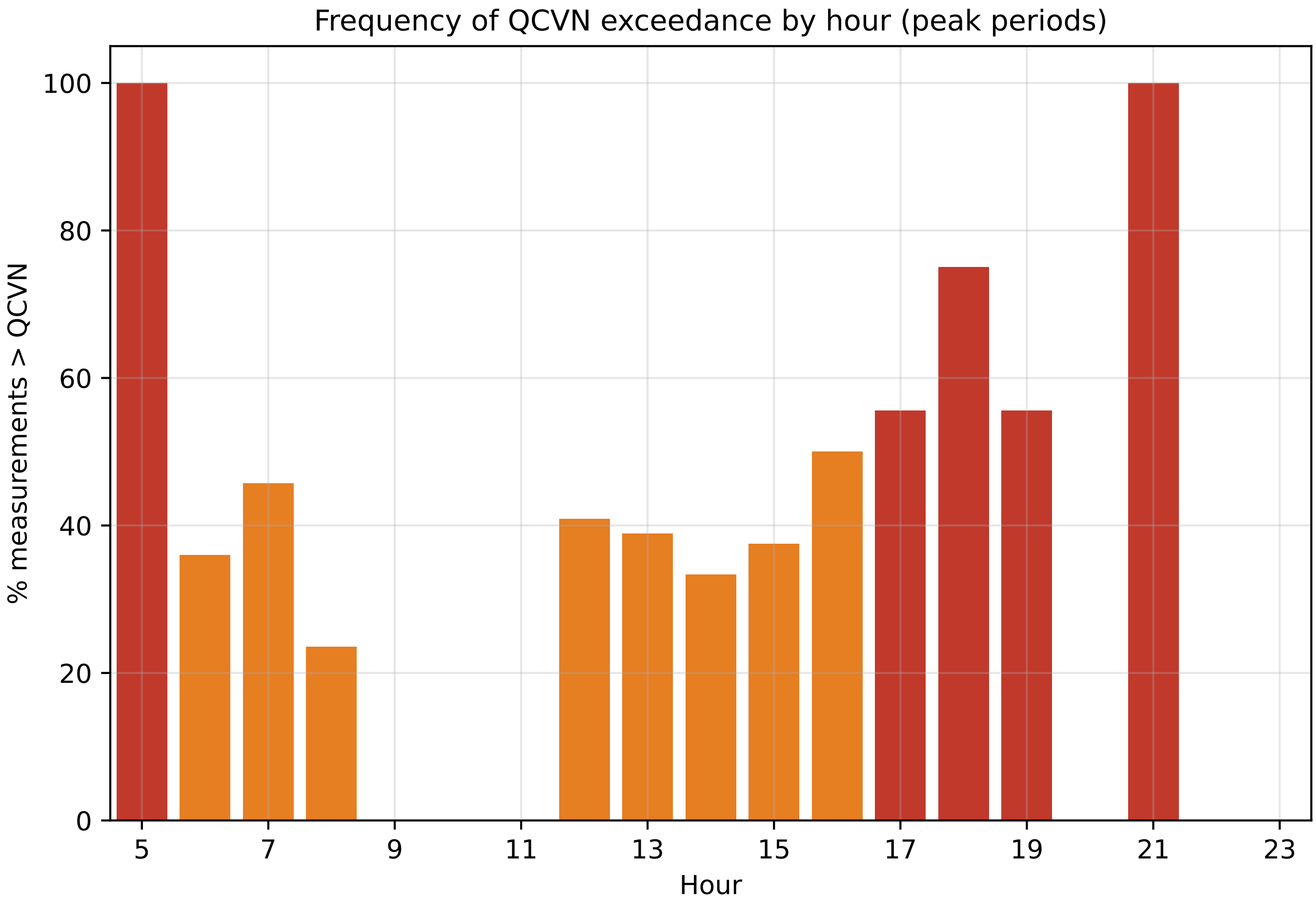




4. Data analysis: sources & exceedances

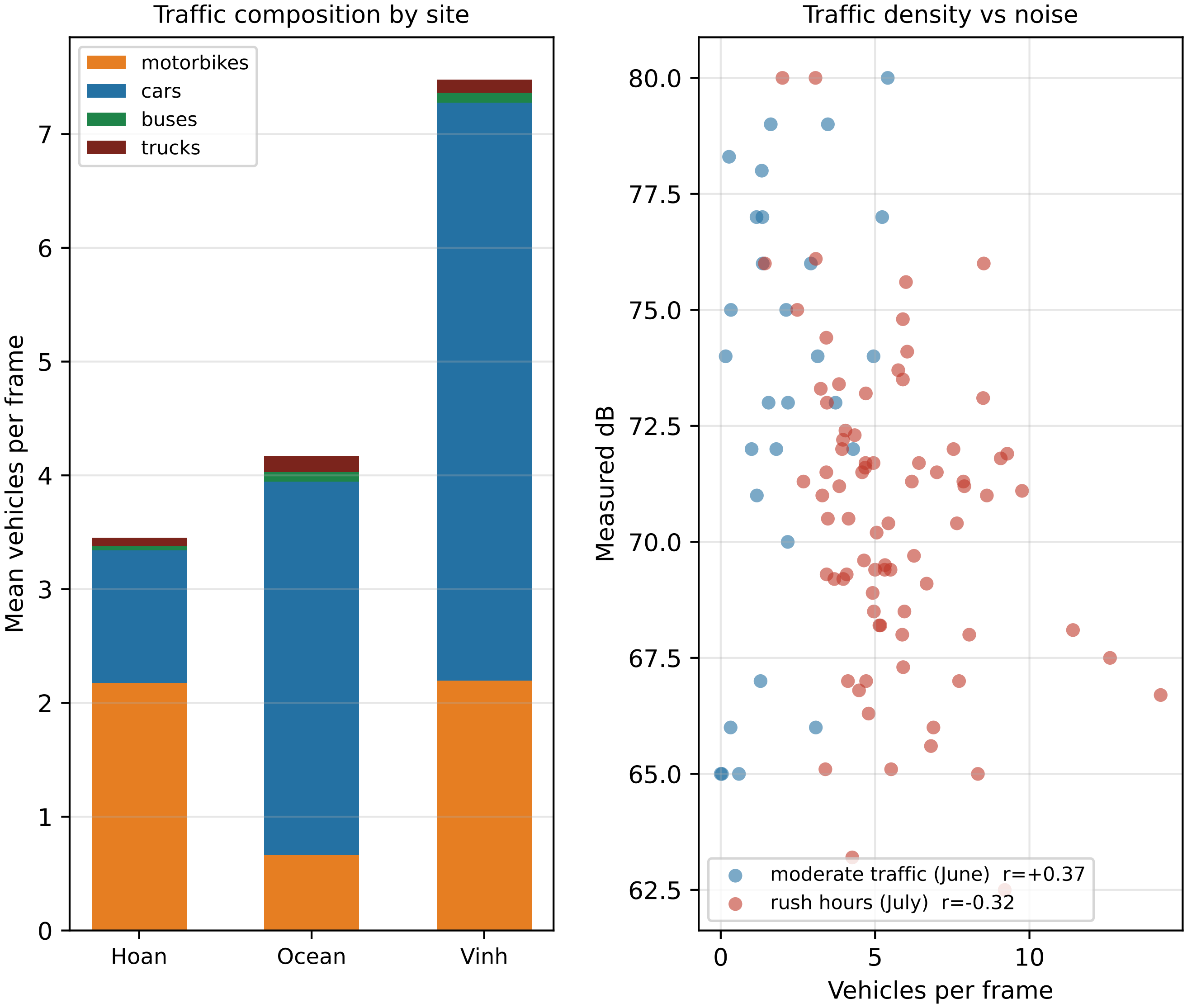






4. Data analysis: traffic videos & weather

107 traffic videos (~25 s each) matched to their measurement by timestamp (median gap 15 s). Vehicles counted with YOLOv8 at ~1 frame/s (2766 frames analysed): average vehicles visible per frame.



Findings: traffic composition matches the zone typologies: Hoan Kiem is motorbike-dominated, Ocean Park car-dominated, Vinh Tuy densest (up to 14 vehicles/frame). In moderate traffic, more vehicles = more noise (r +0.37); at rush hours the correlation reverses (r −0.32): congestion slows traffic down, and slow dense traffic is quieter than free-flowing traffic. Note: small motorbikes are harder to detect than cars, so motorbike shares are likely underestimated.

Weather: no robust effect. Raw correlations (temperature +0.26, rain −0.29) vanish once hour of day and session are controlled: quiet-point campaigns happened to fall on rainy days. Wind shows no effect on readings (no microphone artefact).

5. Predictive model

LightGBM mapping urban morphology + time of day to noise level (dB).

5.1 Key finding: transfer fails, direct training works

The Uganda-pretrained model transfers poorly to Hanoi even after offset calibration ($R^2 < 0$): the noise-morphology relationship learned in Kampala does not hold here, as in the Barcelona cross-city experiment. Training directly on our own measurements works, and is our method.

Method	r	R^2	MAE (dB)
Direct training, spatial CV (honest)	0.68	0.45	4.4
Direct training, random split (optimistic)	0.71	0.49	4.1
Uganda to Hanoi transfer (comparison)	0.06	−1.48	8.4
Barcelona reference (for context)	0.66	0.61	n/a

Spatial CV: tested on locations the model never saw in training (the honest number). Random split: test points can sit metres from training points, which inflates scores; kept as an optimistic upper bound.

5.2 Per-site generalization (leave-one-site-out)

Test site	R^2	Comment
Ocean Park	0.28	best case (largest, most varied sample)
Vinh Tuy	−0.71	distinct transport-corridor typology
Hoan Kiem	−0.28	distinct old-quarter morphology

A stress test: each site is predicted by a model trained on the other two only. Per-site R^2 on such small samples is noisy; the spatial CV above is the reliable number.

Reading the metrics: r = does the model rank places correctly; R^2 = are predicted dB values accurate. Direct training delivers both: r 0.68, R^2 0.45.

6. Limitations

- The model interpolates well within the sampled areas but extrapolates poorly to a district it has never seen: city-wide prediction would need more urban typologies covered.
- Instantaneous readings carry ±5 dB of irreducible noise (a passing bus), capping the achievable R^2 near 0.6; repeated measurements at fixed points would raise it.
- Consumer sound meters: a constant calibration bias is correctable, clipping above 90 dB and wind noise are not. Weekend coverage is still light compared to weekdays.

7. Next steps

- Add weekend sessions and repeated measurements at fixed locations to average out instantaneous variation.
- Vehicle counting from the 83 traffic videos via computer vision, each matched to its measurement point (transport composition as a model feature).
- GAMA simulation: import the noise map, add a traffic-volume slider and what-if scenarios (pedestrianisation, peak-hour traffic).
- Manuscript: methods (Sunbird reproduction + transferability study), results, discussion.

8. Summary

323 calibrated measurements across 3 districts confirm high exposure (39% exceed QCVN). A model trained on our data reaches R^2 0.45 / r 0.68 / MAE 4.4 dB (spatial CV), solid for instantaneous smartphone data. Cross-city transfer from Uganda fails: a documented, useful methodological result.